

Fall 2025, EE 582: Assignment 4:

Induction Machine Modeling

Due on Sunday, November 2nd (submission in Canvas)

SECTION A (CCPD model)

Develop a coupled-circuit phase-domain model in Simulink (using MATLAB functions) for the 500-hp induction machine (IM) from the reference book (p. 165) [provided in the table below]. Try to use a MATLAB script file to store and load model parameters. Also implement a balanced 3-phase sinusoidal voltage source (VS) with variable amplitude to supply the IM with the rated (nominal) voltages. Choose an ode solver with the settings (error tolerances and step-size limits) that you think produces trustable results and is numerically efficient. Comment on your choice.

Table 4.10-1 Induction Machine Parameters

Machine Rating			T_B	$I_{B(abc)}$	r_s	X_{ls}	X_M	X'_{lr}	r'_r	J
hp	Volts	rpm	(N · m)	(amps)	(ohms)	(ohms)	(ohms)	(ohms)	(ohms)	(kg · m ²)
500	2300	1773	1.98×10^3	93.6	0.262	1.206	54.02	1.206	0.187	11.06

Assume a short-circuited (squirrel-cage) rotor. Simulate the start-up transient assuming zero initial conditions and no mechanical load. Run the model up to 2 sec (roughly to steady state). Plot variables v_{qds} , i_{qds} , i_{qdr} , i_{as} , T_e , w_{rm} (all on one page using the subplot command).

SECTION B (qd0 model)

Develop a $qd0$ model in Simulink for the same machine based on the Arbitrary Reference Frame (ARF) so that you can change the frame speed ω . Use different speed for the reference frame: (a) stationary reference frame $\omega = 0$; (b) rotor reference frame $\omega = \omega_r$; and (c) synchronous reference frame $\omega = \omega_e = \omega_b$. In each case, perform the start-up simulation, and plot variables v_{qds} , i_{qds} , i_{qdr} , i_{as} , T_e , w_{rm} (all on one page using the subplot command). Comment on the frequency of the currents and voltages as viewed in different frames of reference. Which reference frame results in the most numerically efficient solution? How do the results (i_{as} , T_e , w_{rm}) compare with the ones obtained using the CCPD model in Section A?

Now you will do a load change study. Assume the machine operates in steady state at no-load. You need to set the initial condition for the no-load steady state. *Hint*: run the model with no-load until it reaches steady state; save the final states and use them as initial states for the next run. You can access the initial and final state settings from Configuration Parameters in Simulation menu of your Simulink model file. Then, simulate this load change: At $t = 0.1$ s, a constant mechanical load of $T_m = 1980$ Nm is applied to the shaft; and at $t = 0.6$ s the mechanical torque is reversed to $T_m = -1980$ Nm. Run the model to $t = 1.2$ s and plot the stator current i_{as} , electromechanical torque T_e , rotor speed ω_{rm} , input real power P_e and input reactive powers Q_e . Explain the results and the flow of real and reactive power.

SECTION C (steady-state model)

Develop the simplified per-phase steady-state model of the machine as described in the lecture notes. Repeat the start-up transient study and the load-change study performed in Section B using this steady-state model. For each case, plot the variables (i_{as} , T_e , w_{rm}); and superimpose (for each variable) the results of the $qd0$ model obtained in section B for comparison. Comment on the accuracy and numerical efficiency of this model compared to the $qd0$ model.

REPORTING

All reports should be typed. The assignment report should demonstrate the individual's work and include printouts of the model windows, simulation results, as well as answers, discussions, and conclusions wherever appropriate. Use graphical commands such as **hold** and **subplot** for compact and clear presentation of the figures in your report. The discussions and comments should not be long (**avoid writing unnecessarily long reports!**), but they should demonstrate your understanding of the problem.

Please include your MATLAB codes as well as screenshots of your Simulink models as part of your report. Your report should be one single pdf file.

Best of luck!